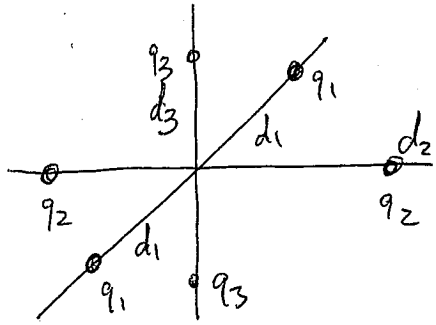
P G6.39



(a) $H' = V_0 + 3(\beta_1 x^2 + \beta_2 y^2 + \beta_3 z^2) - (\beta_1 + \beta_2 + \beta_3) r^2$

$$\beta_i = -\frac{e}{4\pi\varepsilon_0}\frac{q_i}{d_i^3}, \quad V_0 = 2(\beta_1 d_1^2 + \beta_2 d_2^2 + \beta_3 d_3^2)$$

This follows from algebra and I skip the proof.

(b) Correction to the ground state (ψ_{100}) energy

$$\Delta E^1_{100} = \langle \psi_{100} | H' | \psi_{100} \rangle$$

$$\psi_{100} = R_{10}(r)\, Y_0^0(\theta,\varphi) = \frac{1}{\sqrt{\pi a^3}} e^{-r/a}$$

has no angular dependence whatsoever

$\Rightarrow \langle 100 | x^2 | 100 \rangle = \langle 100 | y^2 | 100 \rangle = \langle 100 | z^2 | 100 \rangle$

by symmetry.

Clearly $\langle 100 | r^2 | 100 \rangle = \langle 100 | x^2 + y^2 + z^2 | 100 \rangle =$

$= 3 \langle 100 | x^2 | 100 \rangle$

$$\Rightarrow \langle 100 | H' | 100 \rangle = \langle 100 | V_0 | 100 \rangle +$$

$$+ 3(\beta_1 + \beta_2 + \beta_3) \langle 100 | x^2 | 100 \rangle$$

$$- (\beta_1 + \beta_2 + \beta_3) \underbrace{\langle 100 | r^2 | 100 \rangle}_{3 \langle 100 | x^2 | 100 \rangle} = V_0$$

$\Rightarrow$ correction is simply a constant shift,

$$\Delta E^{(1)}_{100} = V_0$$

(c) Corrections to $\underline{n=2}$ states: ψ_{200}, ψ_{21m} $(m=-1,0,1)$

$$\psi_{200} = R_{20}(r)\, Y_0^0(\theta,\varphi) = \frac{1}{\sqrt{4\pi}} \frac{1}{\sqrt{2a^3}} \left(1-\frac{r}{2a}\right) e^{-r/2a}$$

does not depend on θ, φ

$\Rightarrow$ as in part (b) $\langle 200| H^1 |200\rangle = V_0$

Other diagonal elements:

$$\psi_{21m} = R_{21}(r)\, Y_1^m(\theta,\varphi) = \frac{1}{\sqrt{24\, a^3}} \frac{r}{a} e^{-\frac{r}{2a}} \cdot \sqrt{\frac{3}{4\pi}} \begin{cases} -\sin\theta\, e^{i\varphi} \\ \cos\theta \\ \sin\theta\, e^{-i\varphi} \end{cases}$$

Note that $\langle \psi_{21m} | r^2 | \psi_{21m}\rangle = \langle R_{21} | r^2 | R_{21}\rangle \cdot \underbrace{\langle Y_1^m | Y_1^m\rangle}_{1} =$

$$= \frac{1}{24a^3} \int_0^\infty \left(\frac{r}{a}\right)^2 e^{-r/a}\, r^2 \cdot r^2 dr \underset{\frac{r}{a}=u}{=} \frac{a^2}{24} \underbrace{\int_0^\infty u^6 e^{-u} du}_{6!} = 30a^2.$$

$$\langle \psi_{21m} | x^2 | \psi_{21m}\rangle = \langle \psi_{21m} | y^2 | \psi_{21m}\rangle$$

because both are proportional to $\underbrace{\int_0^{2\pi} \begin{Bmatrix} \cos^2\varphi \\ \sin^2\varphi \end{Bmatrix} d\varphi}_{\pi}$

$\Rightarrow$ express them via

$$\langle 21m | x^2 | 21m\rangle = \frac{1}{2}\left[\langle 21m | r^2 | 21m\rangle - \langle 21m | z^2 | 21m\rangle\right]$$

So $\langle 21m | z^2 | 21m\rangle = \langle R_{21}\, Y_1^m | r^2 \cos^2\theta | R_{21}\, Y_1^m\rangle =$

$$= \underbrace{\langle R_{21} | r^2 | R_{21}\rangle}_{30a^2} \langle Y_1^m | \cos^2\theta | Y_1^m\rangle$$

$$\langle Y_1^{\pm 1} | \cos^2\theta | Y_1^{\pm 1}\rangle = \frac{3}{4\pi} \int \cos^2\theta\, \frac{1}{2} \sin^2\theta \sin\theta\, d\theta\, d\varphi$$

$$= \frac{3}{4} \int_{-1}^{1} \cos^2\theta \left(1-\cos^2\theta\right) d(\cos\theta) = \frac{3}{4} \cdot \frac{4}{15} = \frac{1}{5}$$

Thus $\langle \psi_{21\pm1} | z^2 | \psi_{21\pm1} \rangle = 6a^2 \Rightarrow \langle \psi_{21\pm1} | x^2 | \psi_{21\pm1} \rangle = \frac{30a^2-6a^2}{2} = 12a^2.$

$$\langle Y_1^0 | \cos^2\theta | Y_1^0 \rangle = \frac{3}{4\pi} \int \cos^2\theta \cos^2\theta \sin\theta d\theta d\varphi =$$

$$= \frac{3}{2} \underbrace{\int_{-1}^{1} \cos^4\theta \, d(\cos\theta)}_{\frac{2}{5}} = \frac{3}{5}$$

$$\Rightarrow \langle \psi_{210} | z^2 | \psi_{210} \rangle = 30a^2 \cdot \frac{3}{5} = 18a^2$$

$$\langle \psi_{210} | x^2 | \psi_{210} \rangle = \frac{30a^2 - 18a^2}{2} = 6a^2.$$

$$\langle \psi_{210} | H' | \psi_{210} \rangle = V_0 + 3(\beta_1+\beta_2)\langle x^2 \rangle + 3\beta_3 \langle z^2 \rangle$$

$$- (\beta_1+\beta_2+\beta_3)\langle r^2 \rangle = V_0 + 3(\beta_1+\beta_2)\, 6a^2 +$$

$$+ 3\beta_3 \cdot 18a^2 - (\beta_1+\beta_2+\beta_3)\cdot 30a^2 =$$

$$= V_0 + 12a^2(\beta_1+\beta_2) + 24a^2\beta_3.$$

$$\langle \psi_{21\pm1} | H' | \psi_{21\pm1} \rangle = V_0 + 3(\beta_1+\beta_2)\cdot 12a^2 + 3\beta_3 \cdot 6a^2$$

$$- (\beta_1+\beta_2+\beta_3)\cdot 30a^2 =$$

$$= V_0 + 6a^2(\beta_1+\beta_2) - 12a^2\beta_3.$$

Now we find off-diagonal elements:

V_0 and r^2 can not contribute here because

$$\langle \psi_{21m} | r^2 | \psi_{21m'} \rangle = \langle R_{21} | r^2 | R_{21} \rangle \langle Y_1^m | Y_1^{m'} \rangle = 0$$

Focus on φ-dependence next: $x^2 = r^2 \sin^2\theta \cos^2\varphi$

$y^2 = r^2 \sin^2\theta \sin^2\varphi$

$z^2 = r^2 \cos^2\theta$

$$\begin{cases} \cos^2\varphi = \left(\dfrac{e^{i\varphi}+e^{-i\varphi}}{2}\right)^2 = \dfrac{e^{i2\varphi}+e^{-i2\varphi}+2}{4} \\ \sin^2\varphi = \left(\dfrac{e^{i\varphi}-e^{-i\varphi}}{2i}\right)^2 = \dfrac{e^{i2\varphi}+e^{-i2\varphi}-2}{-4} \end{cases}$$

$\Rightarrow$ can connect states differing by $e^{\pm i2\varphi} \Rightarrow m=+1$ and $m=-1$ states!

$$\langle \psi_{21+1} | x^2 | \psi_{21-1} \rangle = \langle R_{21} | r^2 | R_{21} \rangle \cdot$$

$$\cdot \frac{3}{8\pi} \int \left(-\sin\theta\, e^{i\varphi}\right)^* \sin^2\theta \left(\frac{e^{i2\varphi}+e^{-i2\varphi}+2}{4}\right) \sin\theta\, e^{-i\varphi} \cdot$$

$$\cdot \sin\theta\, d\theta\, d\varphi =$$

$$= 30a^2 \left(\frac{3}{8\pi}\right)(-1) \underbrace{\int_0^\pi \sin^4\theta \sin\theta\, d\theta}_{\frac{16}{15}} \underbrace{\int_0^{2\pi} e^{-i\varphi} \frac{e^{i2\varphi}}{4} e^{-i\varphi} d\varphi}_{\frac{\pi}{2}}$$

$$= -30a^2 \cdot \frac{16}{15} \cdot \frac{3}{8\pi} \cdot \frac{\pi}{2} = -6a^2.$$

Clearly $\langle \psi_{21+1} | y^2 | \psi_{21-1} \rangle = +6a^2$ due to overall $(-)$ sign.

One more remaining possibility: $\langle \psi_{200} | x^2 | \psi_{210} \rangle =$

$$= \langle R_{20} | r^2 | R_{21} \rangle \underbrace{\langle Y_0^0 | \sin^2\theta \cos^2\varphi | Y_1^0 \rangle}_{\int_0^\pi \sin^2\theta \cos\theta \sin\theta\, d\theta = 0 \text{ by oddness}}$$

Thus perturbation matrix is $\langle H' \rangle$

	200	210	211	21-1
200	V_0	0	0	0
210	0	$V_0 + 12a^2(\beta_1+\beta_2) + 24a^2\beta_3$	0	0
211	0	0	$V_0 + 6a^2(\beta_1+\beta_2) - 12a^2\beta_3$	$-18a^2(\beta_1-\beta_2)$
21-1	0	0	$-18a^2(\beta_1-\beta_2)$	$V_0 + 6a^2(\beta_1+\beta_2) - 12a^2\beta_3$

Eigenvalues $\varepsilon_1 = V_0$

$$\varepsilon_2 = V_0 - 12a^2(\beta_1+\beta_2) + 24a^2\beta_3$$

$$\varepsilon_{3,4} = V_0 + 6a^2(\beta_1+\beta_2) - 12a^2\beta_3 \pm 18a^2(\beta_1-\beta_2) =$$

$$= \begin{cases} V_0 + 24a^2\beta_1 - 12a^2\beta_2 - 12a^2\beta_3 \\ V_0 - 12a^2\beta_1 + 24a^2\beta_2 - 12a^2\beta_3 \end{cases}$$

So that finally:

(i) cubic symmetry $(\beta_1 = \beta_2 = \beta_3)$: one level (4-fold degenerate)

(ii) tetragonal $(\beta_1 = \beta_2 \neq \beta_3)$: $\begin{cases} \varepsilon_1 = V_0 \\ \varepsilon_2 = V_0 - 24a^2(\beta_1 - \beta_3) \\ \varepsilon_{3,4} = V_0 + 12a^2(\beta_1 - \beta_3) \end{cases}$

<u>3 levels</u>

(iii) orthorombic $(\beta_1 \neq \beta_2 \neq \beta_3)$: all 4 levels have different energies

$\Rightarrow$ no remaining degeneracy!